Pre-commissioning and commissioning checklist

Flash drum D-400 unit · test pressures, nitrogen inerting, checklist by system, SIS FAT / SAT, start-up, performance test

Document no. FD-V1-CM-PCC-001

What must be done and proven, system by system, between mechanical completion and handover: pressure and leak tests, flushing, loop checks, SIS tests of every C&E row, inerting with nitrogen, the start-up sequence (the OTS start-up procedure) and the performance test against the HMB design case.

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Holds and engineering queries

- EC-02 Test-pack boundaries at the spec breaks
- EC-03 Inerting vent route
- EC-04 Performance test needs good analysers

Note. Numbers come from the simulator model (design case and steady model runs) and the stated assumptions; generated for teaching, not for construction. Engineering queries raised by this document are listed in it and in FD-V1-PM-DOC-001.

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FD-V1-CM-PCC-001	0	Issued for teaching	Apinan Soottitantawat	28-OCT-26

1 · Commissioning systems

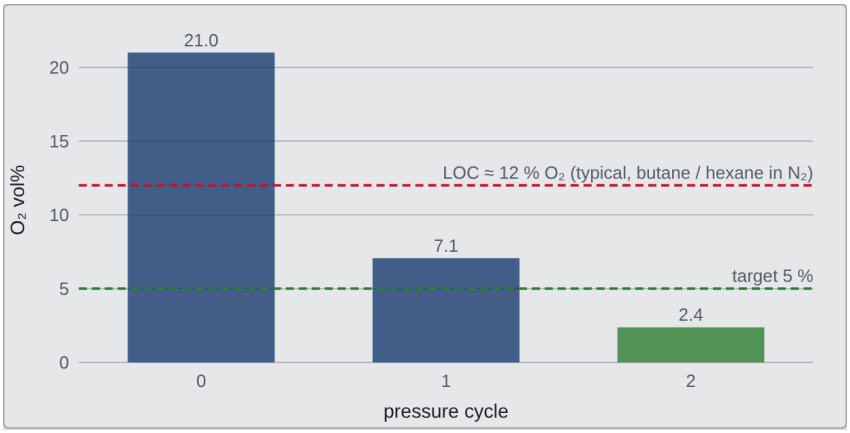
System	Name	Lines	Equipment	Instruments	Checklist items
SYS-01	Feed and let-down	2"-P-4001-A1A, 2"-P-4002-A1A, 6"-P-4004-A1A	E-400 tubes, FCV-400, XV-408	FT-400, TT-401, FIC-400, TIC-401	PC-01, PC-02, PC-03, PC-04, PC-05
SYS-02	Hot oil	2"-HO-4011-B1A, 2"-HO-4012-B1A, HO-4013/4014/4015	E-400 shell, TCV-401, XV-409	FT-405, TI-402, TI-403, PDI-406	PC-06, PC-07, PC-08, PC-09
SYS-03	D-400 and vapour	4"-PV-4021-A1A, 3"-PV-4023-A1A	D-400, PCV-420, PSV-400	LT-400, LG-400, PT-420, FI-420, AI-420, AI-400, LSHH-416, PSHH-426	PC-10, PC-11, PC-12, PC-13, PC-14, PC-15, PC-16
SYS-04	Bottoms	2"-P-4003-A1A, 4"-P-4005-A1A, 2"-CD-4051-A1A	LCV-400, XV-418	FT-410, AI-410, TI-410, LSLL-417	PC-17, PC-18, PC-19
SYS-05	Nitrogen blanket	1"-N-4031-A1A	PCV-430	AI-400	PC-20
SYS-06	Relief and flare	3"-PV-4023-A1A, 6"-FL-4041-F1A	PSV-400	—	PC-21
SYS-07	Instrumentation and SIS	—	DCS, SIS	all loops, I-1, I-2, HS-400	PC-22, PC-23, PC-24, PC-25, PC-26

2 · Test pressures

Test	Design P kPa g	Test P kPa g	Basis
Feed system (to the FCV-400 inlet)	820.0	1,230.0	1.5 × design (ASME B31.3 §345.4.2)
Hot feed line	820.0	1,230.0	1.5 × design (ASME B31.3 §345.4.2)
Drum-side piping (vapour, bottoms, N_2 after PCV-430)	450.0	675.0	1.5 × design (ASME B31.3 §345.4.2)
Hot oil	1,000.0	1,500.0	1.5 × design (ASME B31.3 §345.4.2)
N_2 upstream of PCV-430	1,000.0	1,500.0	1.5 × design (ASME B31.3 §345.4.2)
Flare tail pipe	350.0	525.0	1.5 × design (ASME B31.3 §345.4.2)
D-400 (shop)	450.0	585 top / 640 bottom	UG-99(b)
E-400 shell / tubes (shop)	1000 / 820	1300 / 1066	UG-99(b)
Unit service leak test with N_2	operating 102.7	405	0.9 × PSV set, 1 h, joints snooped
PSV-400 set check	450.0	450 ± 13.8	UG-126(d)

3 · Nitrogen inerting

Pressure cycles with good mixing: $y_n = y_0(P_{L,a}/P_{H,a})^n$. Volume 10.955 m³ (D-400 10.724 m³ + E-400 tubes + piping 0.15 m³ assumed), 0 → 200 kPa g, ratio 0.336: **2 cycles** take O_2 from 21 % to 2.37 % (target 5 %, below the LOC ≈ 12 % of butane / hexane, typical). N_2 19.6 Nm³ per cycle, **39 Nm³** in all. A continuous sweep (mixing factor 2, assumed) would need 29 Nm³. P_H stays below the PSHH-426 trip (250 kPa g). The blanket regulator PCV-430 closes at 30 kPa g: the cycles need a temporary N_2 connection, and the vent must go to the flare (EC-03).



4 · Checklist

Item	System	Activity	Acceptance criterion	Reference	By	Done
PC-01	SYS-01	Flush the feed lines 2"-P-4001-A1A and 2"-P-4002-A1A with clean water, E-400 tubes included (FCV-400 and the orifice FE-400 removed or spooled out)	water clear at the outlet; velocity ≥ the operating 2 m/s (typical)	FD-V1-PE-LIN-001	Commissioning	☐
PC-02	SYS-01	Hydrotest the feed system (test pack to the FCV-400 inlet)	1.5 × 820 = 1230 kPa g, 10 min hold, no leak (ASME B31.3 §345.4.2)	FD-V1-PE-LIN-001	Construction	☐
PC-03	SYS-01	Stroke FCV-400; check the fail-closed action on loss of signal and air	closes on 0 mA and on air off; stroke time recorded	FD-V1-IN-INL-001	Commissioning	☐
PC-04	SYS-01	XV-408 stroke and fail test from the SIS (XY-408 de-energised)	closes in ≤ 2 s; ZSO / ZSC change state; seat tight	SRS FD-V1-IN-SRS-001	Commissioning	☐
PC-05	SYS-01	Loop checks FT-400 / FIC-400, TT-400, TT-401 / TIC-401	DP 0 / 25 / 100 % → 0 / 12.5 / 25 m³/h on FIC-400 (square root); RTD at two points	FD-V1-IN-INL-001 part 4	Commissioning	☐
PC-06	SYS-02	Clean the hot-oil lines without water (blow with dry air, then circulate oil through a temporary strainer)	no water left: it would flash at 180 °C; strainer clean after 4 h	FD-V1-PE-LIN-001	Commissioning	☐
PC-07	SYS-02	Pressure test the hot-oil branch (with oil, or water then dried to a dew point below 0 °C)	1.5 × 1000 = 1500 kPa g	FD-V1-PE-LIN-001	Construction	☐
PC-08	SYS-02	Stroke TCV-401 both ports; fail-to-bypass test; XV-409 stroke and fail test	0 % = all oil on the bypass; XV-409 closes ≤ 2 s	FD-V1-IN-INL-001, SRS	Commissioning	☐
PC-09	SYS-02	Loop checks FT-405 / FIC-405, TT-402, TT-403, PDT-406 / PDI-406	readings match the injected signal ± 0.25 % of span	FD-V1-IN-INL-001	Commissioning	☐
PC-10	SYS-03	D-400 shop hydrotest certificate reviewed (no field hydrotest of the vessel)	585 kPa g at the top (UG-99(b)); 640 kPa g at the bottom if tested vertical	FD-V1-ME-DS-001	QA/QC	☐
PC-11	SYS-03	Inspect the internals before closing the manway M1 (mesh pad, inlet device, vortex breaker)	clean, bolted, mesh pad seated; manway gasket new	FD-V1-ME-DS-001	Mechanical	☐
PC-12	SYS-03	Air-blow the vapour lines 4"-PV-4021-A1A (not water)	target plate clean	FD-V1-PE-LIN-001	Commissioning	☐
PC-13	SYS-03	Pressure test the drum-side piping (vapour, bottoms, N_2 downstream of PCV-430), D-400 blinded off or included at its own test pressure	1.5 × 450 = 675 kPa g on the piping; D-400 not above 585 kPa g	FD-V1-PE-LIN-001	Construction	☐
PC-14	SYS-03	LT-400 calibration with the remote seals (LRV / URV from IN-INL-001 part 4); LG-400 against LT-400 on water	LRV −29.22 kPa, URV −11.55 kPa (service density 605 kg/m³)	FD-V1-IN-INL-001 part 4	Commissioning	☐
PC-15	SYS-03	Loop checks PT-420 / PIC-420, FT-420, AT-420, AT-400; PCV-420 stroke and fail-open test	PCV-420 opens on 0 mA and on air off	FD-V1-IN-INL-001	Commissioning	☐
PC-16	SYS-03	PSV-400 set-pressure certificate and tag checked; inlet and outlet clear	set 450 ± 13.8 kPa g (UG-126(d)); orifice per RLB-001	FD-V1-PE-RLB-001	QA/QC	☐
PC-17	SYS-04	Flush the bottoms lines 2"-P-4003-A1A and 4"-P-4005-A1A; LCV-400 spooled out	water clear	FD-V1-PE-LIN-001	Commissioning	☐
PC-18	SYS-04	Stroke LCV-400 (fail closed) and XV-418 (SIS, fail closed)	both close on loss of signal / de-energise	FD-V1-IN-INL-001	Commissioning	☐
PC-19	SYS-04	Loop checks FT-410 / FIC-410, AT-410 (with SC-410), TT-410 / TI-410	as above	FD-V1-IN-INL-001	Commissioning	☐
PC-20	SYS-05	Blow the N_2 line with N_2; set PCV-430 on the bench	set 30 kPa g; restriction orifice installed (ER-06)	FD-V1-PE-RLB-001	Commissioning	☐
PC-21	SYS-06	Check the PSV tail pipe 6"-FL-4041-F1A: free-draining to the flare header, no pockets, supports	slope toward the header; no low point	FD-V1-PE-RLB-001	Mechanical	☐
PC-22	SYS-07	Loop check every instrument loop of the INL (21 loops) from the field to the DCS / SIS graphic	every loop signed off; alarms at the ALM settings	FD-V1-IN-INL-001	Commissioning	☐
PC-23	SYS-07	SIS site acceptance test: every C&E row tested end to end (sensor → logic → XV)	all 27 rows pass (see FAT / SAT table)	C&E FD-V1-IN-CE-001	Instrument / safety	☐
PC-24	SYS-07	SIS proof test before first start-up (the SRS test procedure)	trip settings 90 %, 250 kPa g, 5 %; delays 5 / 3 / 5 s; latch and reset as specified	SRS FD-V1-IN-SRS-001	Instrument / safety	☐
PC-25	SYS-07	UPS autonomy test with the DCS and SIS on load	≥ 30 min (assumed)	FD-V1-EE-LL-001	Electrical	☐
PC-26	SYS-07	Initial Ex inspection of every field item and earthing / bonding check	IEC 60079-17 detailed inspection; IIA T3 or better	FD-V1-EE-LL-001	Electrical	☐
PC-27	Unit	Leak test the unit with N_2 (D-400 system)	405 kPa g (0.9 × PSV set), 1 h hold, all joints snooped, no bubbles	this document	Commissioning	☐
PC-28	Unit	Inert with N_2 by pressure cycles to remove the air	2 cycles 0 → 200 kPa g give O_2 2.4 % < 5 %; 39 Nm³ N_2; vent to the flare	part 6 calculation	Commissioning	☐
PC-29	Unit	Leave D-400 under the N_2 blanket (HV-430 open, PCV-430 30 kPa g) until feed is introduced	PI-420 ≈ 30 kPa g; AI-400 reads ≈ 100 %	start-up procedure	Operations	☐
PC-30	Unit	Mechanical completion certificate and punch list A cleared; operating procedures and the C&E issued	signed	—	Construction / operations	☐

5 · SIS FAT / SAT: every row of FD-V1-IN-CE-001

Row	Cause	Test method	Expected (C&E)	Result on a copy of the model	Pass
1	**HS-400** Emergency shutdown push-button (DCS / SIS console)	ESD push-button pressed	close XV-408, XV-409, XV-418; delay 0 s	closed XV-408, XV-409, XV-418; after 0.00 s	✓
2	**LSHH-416** D-400 level high-high: liquid toward the vapour line (own level switch)	simulated high level (trip setting moved below the 50 % level)	close XV-408, XV-409; delay 5 s	closed XV-408, XV-409; after 5.25 s	✓
3	**PSHH-426** D-400 pressure high-high (own pressure switch)	simulated high pressure (trip setting moved below 102.7 kPa g)	close XV-408, XV-409; delay 3 s	closed XV-408, XV-409; after 3.25 s	✓
4	**LSLL-417** D-400 level low-low: gas blow-by to the liquid line (own level switch)	simulated low-low level with XV-418 open (setting moved above the level)	close XV-418; delay 5 s	closed XV-418; after 5.25 s	✓
5	**FIC-400** Feed flow low (FT-400)	signal injected past the setting	Low 5 m³/h: alarm + horn	ALM 5	✓
6	**FIC-400** Feed flow high (FT-400)	signal injected past the setting	High 20 m³/h: alarm + horn	ALM 20	✓
7	**TIC-401** Hot feed temperature low (TT-401)	signal injected past the setting	Low 69 °C: alarm + horn	ALM 69	✓
8	**TIC-401** Hot feed temperature high (TT-401)	signal injected past the setting	High 79 °C: alarm + horn	ALM 79	✓
9	**LIC-400** D-400 level low (LT-400)	signal injected past the setting	Low 20 % span: alarm + horn	ALM 20	✓
10	**LIC-400** D-400 level high (LT-400)	signal injected past the setting	High 80 % span: alarm + horn	ALM 80	✓
11	**PIC-420** D-400 pressure low (PT-420)	signal injected past the setting	Low 70 kPa g: alarm + horn	ALM 70	✓
12	**PIC-420** D-400 pressure high (PT-420)	signal injected past the setting	High 130 kPa g: alarm + horn	ALM 130	✓
13	**FI-420** Overhead vapour flow low	signal injected past the setting	Low 300 Nm³/h: alarm + horn	ALM 300	✓
14	**FI-420** Overhead vapour flow high	signal injected past the setting	High 900 Nm³/h: alarm + horn	ALM 900	✓
15	**FIC-410** Bottoms liquid flow low	signal injected past the setting	Low 8 m³/h: alarm + horn	ALM 8	✓
16	**FIC-410** Bottoms liquid flow high	signal injected past the setting	High 18 m³/h: alarm + horn	ALM 18	✓
17	**AI-410** Bottoms butane above the 28 wt% spec	signal injected past the setting	High 28 wt% C4: alarm + horn	ALM 28	✓
18	**AI-420** Overhead butane below the 75 wt% spec	signal injected past the setting	Low 75 wt% C4: alarm + horn	ALM 75	✓
19	**AI-400** N2 in the drum vapour high	signal injected past the setting	High 5 % N2: alarm + horn	ALM 5	✓
20	**TI-410** Flash temperature low (drum liquid)	signal injected past the setting	Low 40 °C: alarm + horn	ALM 40	✓
21	**TI-410** Flash temperature high (drum liquid)	signal injected past the setting	High 60 °C: alarm + horn	ALM 60	✓
22	**PDI-406** Hot-oil header ΔP low	signal injected past the setting	Low 100 kPa: alarm + horn	ALM 100	✓
23	**PDI-406** Hot-oil header ΔP high	signal injected past the setting	High 200 kPa: alarm + horn	ALM 200	✓
24	**PSV-400** Relief valve lifting to flare (set at the MAWP)	event alarm, simulated at SAT	Lifting 450 kPa g: alarm + horn	configured	✓
25	**D-400** Liquid carried over into the vapour line	event alarm, simulated at SAT	Carry-over — —: alarm + horn	configured	✓
26	**SIS-RST** I-1 reset permissive: level < 85 % and pressure < 230 kPa g	I-1 tripped, cause cleared; open XV-408 before reset, then reset at the console; I-1 tripped with the level above 85 % (liquid added to the copy)	XV-408 blocked until reset; reset blocked above 85 %	"Open blocked: I-1 trip latched. Reset the SIS first."; reset accepted; level 93 %: "Reset blocked: D-400 level still above 85 % (lower it first)."	✓
27	**XV-418** XV-418 open permissive: level above 10 % and I-2 reset	I-2 latched, level below 10 % (liquid removed from the copy); try to open XV-418	XV-418 cannot be opened	level −11 %: "Open blocked: LSLL-417 (level below 10 % or not reset)."	✓

6 · Start-up mapped to the OTS procedure

Step	OTS start-up step (Cold startup)	Commissioning meaning	Complete before
1	Outside: block valves HV-401, HV-421 and HV-411 open	Line-up after the leak test and inerting	PC: leak test, inerting, block valves checked
2	SIS reset; XV-408 (feed) and XV-409 (hot oil) open	SIS reset; first opening of the SIS valves on process	SIS SAT and proof test passed
3	PIC-420 in AUTO with set point about 70 % of design (60–85 kPa g)	Pressure control on the blanket and the vapour header	PCV-420 loop check, PIC-420 alarms
4	FIC-400 in AUTO at about half of design (7–9 m³/h; below about 6 m³/h TIC-401 cycles)	Introduce feed at half rate (first hydrocarbon in the unit)	feed system flushed, tested; FT-400 loop checked
5	TIC-401 in MAN; TCV-401 opened by hand until the hot feed is above 60 °C	Heat the feed slowly by hand (thermal shock, leaks at temperature)	hot-oil system dry and tested; TCV-401 stroked
6	TIC-401 set point 74 °C and in AUTO	Close the temperature loop	TT-401 loop checked
7	N2 swept out of the drum vapour (AI-400 below 2 %)	Sweep the nitrogen to the header (AI-400 falling)	AT-400 calibrated; vapour header ready to receive
8	Level above 15 %: SIS I-2 reset and XV-418 open	First liquid to the downstream unit; I-2 reset	LCV-400, XV-418 stroked; downstream unit ready
9	LIC-400 in AUTO at 50 %	Close the level loop	LT-400 calibrated with the service density
10	PIC-420 set point raised to 102.7 kPa g	Raise the pressure to design	PSHH-426 tested; PSV-400 certificate
11	FIC-400 raised in steps to 16 m³/h	Raise the feed to design in steps	line-out; analysers warm and calibrated
12	Lined out for 10 min: TIC 72–76 °C, PIC 97–108 kPa g, LIC 40–60 %, AI-410 ≤ 28, AI-420 ≥ 75 wt%	Line out: start of the performance test window	see part 9

7 · Performance test

Guarantee	Criterion	Design case (HMB)	Note
Feed rate FI-400	16.00 ± 1 % m³/h	16.00	design throughput
Hot-feed temperature TI-401	74.0 ± 1 °C	74.00	
D-400 pressure PI-420	102.70 ± 3 % kPa g	102.70	
Flash temperature TI-410	49.2 ± 1 °C	49.22	
Vapour product FI-420	611.00 ± 5 % Nm³/h	611.00	vapour make
Vapour butane AI-420	≥ 75 wt% C4	76.82	product guarantee ≥ 75 wt%
Bottoms flow FI-410	14.03 ± 3 % m³/h	14.03	
Bottoms butane AI-410	≤ 28 wt% C4	26.56	product guarantee ≤ 28 wt%
Hot oil to E-400 FI-405	≤ 11.53 t/h	10.98	utility consumption ≤ design + 5 %

Window 24–72 h at design feed, steady, analysers calibrated with certified standards and checked against laboratory samples (EC-04); tolerances typical.

8 · Mechanical completion and punch list

Category	Must be cleared	Typical items
A	Must be cleared before pre-commissioning or before introducing hydrocarbons	safety, pressure integrity, SIS, relief
B	Must be cleared before handover / end of the performance test	function works, but incomplete or a nuisance
C	May be cleared after handover	cosmetic, documentation, spares

Example item	Category	Why
PSV-400 installed without its set-pressure certificate	A	a relief device whose set is not proven cannot protect D-400
XV-418 closed limit switch not adjusted (no "closed" indication)	A	the SIS cannot confirm the valve closed: SIS function not proven
Drain HV-415 open-ended, no blind or plug fitted	A	open drain on a hydrocarbon system = release on start-up
Spectacle blind at the N_2 connection left in the closed position	A	D-400 cannot be blanketed or purged
Insulation cladding of 2"-HO-4011-B1A not sealed at one elbow	B	water ingress and burns risk; must be done before handover, not before start-up
LG-400 scale plate missing	B	the gauge still works on the float; fix before handover
AT-420 sample-line tracing thermostat label missing	B	function fine; documentation for handover
Paint touch-up at the E-400 saddle bolts	C	cosmetic; after handover
Nameplate of FT-410 faces the wall	C	cosmetic
Spare gasket set for M1 not delivered to the store	C	operations spares; after handover

9 · Engineering queries

No.	Subject	Query	Why it matters	Proposed answer	Status
EC-02	Test-pack boundaries at the spec breaks	The feed system tests at 1230 kPa g (1.5 × 820) and the hot-oil branch at 1500 kPa g, but D-400 may see no more than its shop test 585 kPa g. FCV-400, LCV-400 and the XVs sit on the breaks.	A test pack that includes D-400 by mistake overpressures the vessel; control valves are not test blinds.	Make test packs with blinds at FCV-400 inlet, LCV-400 outlet and the N_2 regulator; remove or spool out control valves and orifice plates during flushing and testing.	OPEN
EC-03	Inerting vent route	The N_2 pressure cycles (2 × 0 → 200 kPa g) must vent air / N_2 somewhere. D-400 has no vent connection (EM-05); through PCV-420 the mixture would go to the fuel-gas (vapour) header.	Air into a fuel-gas header makes a flammable mixture in the header and at the users.	Use the proposed vent V1 to the flare (or a temporary vent to a safe location) for the inerting; keep HV-421 closed until AI-400 shows the drum free of air.	OPEN

No.	Subject	Query	Why it matters	Proposed answer	Status
EC-04	Performance test needs good analysers	The guarantees are AI-420 ≥ 75 wt% (design 76.8, margin 1.8) and AI-410 ≤ 28 wt% (design 26.6, margin 1.4). The margins are of the order of a GC's calibration error.	An uncalibrated or condensing analyser can fail or pass the test on its own error.	Calibrate AT-410 / AT-420 with certified standards before the test, confirm the sample-line tracing (EP-09, EE-04) and take laboratory samples in parallel.	OPEN